

AI Image Generator

Generate images from text prompts using **Stable Diffusion** models (v2.1 and v3.5) via the HuggingFace `diffusers` library.

· Features

- **Text-to-image generation** with customizable prompts
- **Two model versions** · Stable Diffusion 2.1 and 3.5
- **Batch generation** · Generate multiple images from a list of prompts
- **GPU acceleration** · Automatic CUDA support for fast inference

· File Structure

File	Model	Description	----- ----- -----	2-1.py	Stable Diffusion 2.1	Batch generation with multiple prompts, 768×768	3-5.py	Stable Diffusion 3.5	Single-prompt generation, 512×512
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· Getting Started

Prerequisites

- Python 3.10+, NVIDIA GPU with CUDA (recommended)

Installation

```
cd AI-Image-Generator
pip install -r requirements.txt
```

Generate Images

```
python 2-1.py      # Stable Diffusion 2.1 (batch)
python 3-5.py      # Stable Diffusion 3.5
```

Images saved as `image_0.png`, `image_1.png`, etc.

· Logic Flow

- 1 Load `StableDiffusionPipeline` from HuggingFace
- 2 Move pipeline to CUDA device
- 3 Run pipeline with text prompts and inference parameters
- 4 Save generated images as PNG files

Key Parameters

Parameter	2-1.py	3-5.py	Purpose	----- ----- ----- -----	num_inference_steps	50	28		
Denoising iterations		guidance_scale	7.5	3.5	Prompt adherence		height × width	768×768	512×512

· Dependencies

`diffusers[torch], huggingface_hub, transformers, accelerate, sentencepiece, protobuf`

· License

Educational project · use freely for learning and reference.